

Not Just a Gimmick: A Preliminary Study on Designing Interactive Media Art to Empower Embedded Culture's Practitioner

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Abstract

Interactive Media Art (IMA) offers a promising avenue for preserving and promoting cultural heritage by engaging audiences through immersive experiences. However, its potential to support the creation of cultural content and empower practitioners has been largely underutilized. Taking shadow puppetry, a traditional art form often featured in IMA, as a cut point to examine how current IMA practices can be reimagined to better serve the needs of cultural custodians, this paper adopting a practitioner-centered design approach, aim to develop IMA content that not only showcases traditional art but also supports its sustainable innovation and development. While our research is still in its early stages, short-term experiments suggest that IMA can serve as an effective tool for creative support, assisting artists in producing their work. The long-term impact of these systems on artistic output remains to be explored.

CCS Concepts

• Human-centered computing; • Interaction design; • Interaction design process and methods;

Keywords

Interactive Media Art, Cultural Heritage Preservation, User Experience Study

ACM Reference Format:

Yihao He. 2024. Not Just a Gimmick: A Preliminary Study on Designing Interactive Media Art to Empower Embedded Culture's Practitioner. In *SIGGRAPH Asia 2024 Posters (SA Posters '24)*, December 03–06, 2024, Tokyo, Japan. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3681756.3697872>

1 Introduction

Interactive Media Art (IMA) has emerged as a powerful tool in cultural heritage preservation, celebrated for its ability to recreate structures and visual styles while engaging audiences with heritage through interactive methods.[Tost and Champion, 2007; Foni et al., 2010] By allowing users to experience cultural artifacts firsthand,

IMA demonstrates how traditional art forms can embrace modern technology.[Giaccardi, 2012]

However, while the use of IMA to promote cultural content is promising, its potential to contribute to the creation of such content and support practitioners remains largely untapped. Many IMAs, though visually appealing, often fail to make a meaningful impact on the preservation and evolution of traditional art, becoming mere "art on the wall." [Champion, 2016; Jones and Yarrow, 2013] Our research highlights this gap, stressing the need for IMA to evolve beyond a "fancy display" and become a tool that genuinely supports the cultural communities it draws from, fostering sustainable innovation and development in traditional art forms.[Warr and O'Neill, 2005] The true significance of IMA lies not just in how it presents traditional art through new media, but in its potential to contribute to the long-term growth and sustainability of these cultural practices.

In this paper, we focus on shadow puppetry, a common cultural theme in IMA design, and assess the current state of IMA design through the lens of the practitioner community. By adopting a new design approach, we aim to create IMA content that supports both the creation of traditional art and technological development. Our preliminary research demonstrates the potential and feasibility of these IMA applications, with future work exploring the long-term impact of IMA on practitioners' creative processes.

2 ShadowTroupe Workshop: Towards a More Sustainable IMA Solution for Practitioners

2.1 Field Research: Misalignment in Needs Between Practitioners and Designers

We began by exploring shadow puppetry, a theme often integrated into IMA and examined the creative experiences of shadow puppet artists. We interviewed three distinct groups of artists, totaling 14 individuals, all of whom had previously collaborated with new media art design teams. Without delving into specific design details, we focused on the roles these artists played, their level of involvement, and their experiences in these new media projects. From a sustainability perspective, we explored the impact of new media art on their practices and sought their views on its influence and application to traditional art forms.

The results indicate that the influence between IMA and traditional art is minimal. IMA outputs are frequently disconnected from their cultural roots, a misalignment often attributed to differing priorities between practitioners and designers. Designers typically prioritize how traditional art is received by a broader audience, which often leads to the sacrifice of traditional art's core essence

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SA Posters '24, December 03–06, 2024, Tokyo, Japan

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ACM ISBN 979-8-4007-1138-1/24/12

<https://doi.org/10.1145/3681756.3697872>

during adaptation to new media formats, such as in the design of interaction methods, to cater to wider user preferences. Consequently, traditional art's visual concepts are conveyed, but its core essence is significantly compromised. For traditional artist communities, many IMA projects aim to address the lack of attention toward traditional art. However, this is only one aspect of the decline of cultural heritage art. The disappearance of the practitioner communities and the long-term stagnation of techniques, which are the root sustainability issues threatening traditional art forms, are rarely addressed by researchers. Moreover, projects that barely touch the surface of influencing the original theme quickly decay due to shifting designer interests and a lack of long-term support, leaving behind an “untouchable new media mess” for practitioners to manage.

2.2 ShadowTroupe Design: IMA as a Creativity Support Tools for Practitioners

Based on the design challenges identified from our interviews, we re-evaluated the creative practices and design approaches of IMA within the context of traditional art. Our goal was to develop IMA that better align with the needs of the artist community. We shifted the typical IMA design approach by considering the perspective of the practitioners, conducting field studies on the creative processes of shadow puppet artists, and establishing a foundational workflow for design practice based on these insights. In collaboration with the artists, we explored various parts of this workflow, presenting IMA interpretations and brainstorming potential enhancements to improve workflow efficiency.

We aimed to use the complete shadow puppet design process as the foundation for the IMA experience, to showcase how the art form is created from ideas to finalization, offering a comprehensive experience of the entire workflow. Currently, we have developed several small systems using Unity and employed on Quest 3, specifically for rapid prototyping (puppet making) and dance choreography. (See Figure 1) We experimented with virtual reality headsets as image capture tools, allowing sketches to be quickly processed and transformed into controllable shadow puppet models. These models were integrated into a realistic, full-body-control, embodied interaction system, designed according to the unique control dynamics of shadow puppetry. (See Figure 2) Artists use this system to simulate dance choreography, test and practice within a pre-set virtual stage environment, and apply the results in design iteration and reflection. To ensure that cultural content was accurately conveyed, we maintained close communication with the artist community throughout the process.

We further empowered the artists by returning the ability to adjust control details, allowing them to tailor the system to their operational habits and closely replicate their real-world creative experiences.

In our initial small-scale usability testing with artists (n=6, 5 male, 1 female), the system performed well in both efficiency and cultural fidelity, with practitioners easily adapting and integrating ShadowTroupe into their existing workflows. Feedback indicated that practitioners recognized this new media as a valuable support tool for their practice. They praised the effort and results in recreating the intricate techniques of the original performances through

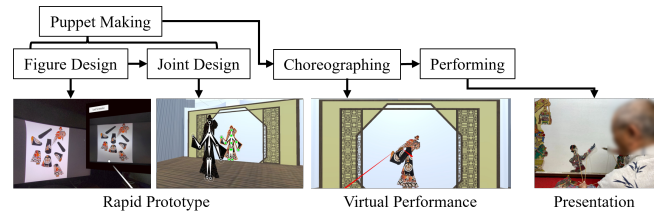


Figure 1: ShadowTroupe's Integration into Basic Shadow Puppet Design Workflow Demonstration.

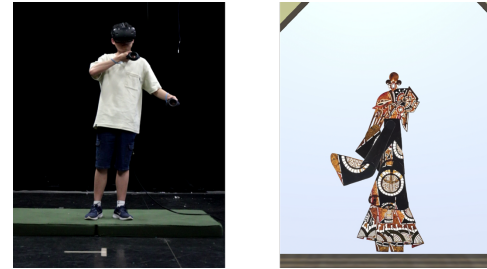


Figure 2: ShadowTroupe's Interaction Demonstration. Artist Utilize This System to Simulate Puppet Control Performance.

IMA design but noted that further refinement is needed in some details. However, we see evaluating the impact of the system on the production of traditional art content as the long-term goal. By the time this paper is written, we have been collaborating with the practitioner community for four months, continuously refining the system through practical application.

3 Conclusion

This paper presents our initial reflections on the integration of new media art within traditional art, explores how IMA design can be incorporated into traditional art design workflows. Future research aims to reveal the long-term impact of new IMA designs on traditional art creation and technological innovation. We hope this shift in design thinking will engage the HCI research and IMA design communities, drawing attention to the needs of practitioners and encouraging sustainable support for cultural Practitioners.

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